Journal of Retailing and Consumer Services 85 (2025) 104282

Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser

AI service may backfire: Reduced service warmth due to service provider transformation

Xingyang Lv [a,b], Yufan Yang [c,d], Dazhi Qin [e], Xiaoyan Liu [f,*]

[a] School of Business Administration, Faculty of Business Administration, Southwestern University of Finance and Economics, 611130, Chengdu, China
[b] Center for Bay of Bengal Studies, Southwestern University of Finance and Economics, 611130, Chengdu, China
[c] College of Tourism and Service Management, Nankai University, 300350, Tianjin, China
[d] School of Business Administration, Southwestern University of Finance and Economics, 611130, Chengdu, China
[e] Faculty of Hospitality and Tourism Management, Macau University of Science and Technology, 999078, Macau
[f] Research Institute of Economics and Management, Southwestern University of Finance and Economics, 611130, Chengdu, China

ARTICLE INFO

Keywords:
AI service
Perceived service warmth
Mind perception
Continuous usage intention
Language style
Customer motivation

ABSTRACT

Perceived service warmth, arising from interactions between customers and employees, is a fundamental component of the hospitality industry. However, as artificial intelligence (AI) facilities such as service robots and chatbots gradually replace the role of human staff, the inherent warmth of the hospitality industry may be under threat. Therefore, the current research aims to investigate how such service provider transformations might influence customer warmth perceptions, the underlying mechanism, and practical solutions for firms that plan to deploy AI facilities in the future. Using five studies (and a supplemental study), we demonstrate that AI (vs. human) service significantly reduces customers' perceived service warmth due to the diminished mind perceptions (i.e., experience and agency) of AI staff as service providers. Such a warmth loss effect further leads to lower customers' continuous usage intention of AI service. However, the effect is attenuated when AI staff interacts with customers using an informal (vs. formal) language style or for customers with a utilitarian (vs. hedonic) motivation. The findings of this research contribute to the literature on AI service and provide valuable insights for marketers into effective AI deployment in the hospitality industry.

1. Introduction

Artificial intelligence (AI) facilities, such as service robots, chatbots, and virtual agents, have been rapidly deployed within the hospitality industry over the past decade (Chi et al., 2020), manifesting in establishments such as hotels (e.g., FlyZoo Hotel in China and Henn-na Hotel in Japan) and restaurants (e.g., Haidilao's robot restaurant in China). The International Federation of Robotics (2022) reports a 37% growth in global sales of service robots in 2022, projecting a market value of US $62.35 billion by 2030. This technology shift is increasingly replacing traditional customer–employee interactions with customer–technology interactions, thus raising concerns about the potential displacement of human employees in the hospitality sector (Huang and Rust, 2018; Yam et al., 2022). Indeed, forecasts predict that AI may replace approximately 47% of human service jobs within two decades (Frey and Osborne, 2017), underscoring its transformative impact.

AI deployment offers benefits such as decreased operational costs, increased efficiency (Shin et al., 2019), enhanced functionality (e.g., speed and convenience), and hedonic benefits (e.g., vividness and enjoyment) for customers (Bilal et al., 2024; Buhalis and Sinarta, 2019; Gursoy et al., 2019). Existing research has predominantly focused on the factors that promote AI acceptance and usage, such as perceived ease of use, perceived usefulness, and anthropomorphism (e.g., Gursoy et al., 2019; Lu et al., 2019; Pham et al., 2024). However, a growing research stream suggests that AI adoption can negatively affect customers' emotions, psychological connections, and trust in service providers because of the inherent human need for interpersonal interactions (e.g., Chi et al., 2020; Granulo et al., 2021; Longoni et al., 2019; Mende et al., 2019; Moore et al., 2022). These negative effects might be even more pronounced in high-touch industries, such as hospitality, where perceived service warmth derived from customer–employee interactions is crucial for customer satisfaction and retention (Gao and Mattila, 2014; Quach et al., 2022).

However, research on how, why, and when replacing human services

* Corresponding author.
E-mail addresses: xingyang.lv@foxmail.com (X. Lv), yufan.yang@foxmail.com (Y. Yang), dazhiqin77@gmail.com (D. Qin), lxy@swufe.edu.cn (X. Liu).

https://doi.org/10.1016/j.jretconser.2025.104282
Received 31 March 2024; Received in revised form 19 November 2024; Accepted 1 March 2025
Available online 18 March 2025
0969-6989/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

with AI may affect customer perceived warmth in the hospitality industry remains limited. Moreover, although the idea that customers perceive less service warmth from AI than from human staff appears to be intuitive, the underlying mechanisms of this relationship require further investigation. Therefore, this study aims to provide a more comprehensive understanding of whether replacing human employees with AI staff, thereby reducing human interaction during service encounters, diminishes customers' perceived service warmth and continuous usage intention, and if so, what forces drive this adverse effect and how service firms mitigate such warmth loss effect in the AI age?

Through five studies (and a supplemental study), we examine the effects of AI (vs. human) service on customer perceived service warmth, uncovering its underlying mechanism, downstream consequences, and boundary conditions. Studies 1 and 2 (and the supplemental study) validate that shifting the service provider from a human to an AI reduced customer perceived warmth across different research methods and samples (both Chinese and Americans). Study 3 identifies diminished mind perception as the primary mechanism underlying lower perceived warmth in AI (vs. human) service contexts. As a coping strategy, Study 4 suggests that using an informal (vs. formal) language style can help to restore the reduced warmth perceptions in AI service encounters. Given that the number of firms deploying AI service keeps growing, Study 5 points out that targeting utilitarian- (vs. hedonic-) oriented customers will result in less negative consequences on customers' behaviors, such as continuous usage intention. Managerially, our findings offer valuable insights for hotel managers regarding when and how to deploy AI facilities while maintaining customer perceived service warmth and continuous usage intentions.

2. Conceptual background

2.1. Warmth in the hospitality industry

Warmth is defined as 'a positive, mild, volatile emotion involving physiological arousal and precipitated by experiencing directly or vicariously a love, family, or friendship relationship' (Aaker et al., 1986, p. 366). Individuals view warmth as being good-natured, tolerant, sincere, trustworthy, and friendly in social contexts, and perceived warmth is a crucial predictor of impression formation and long-term relationships (Fiske et al., 2007; Güntürkün et al., 2020; Min et al., 2023). In the hospitality industry, perceived warmth refers to customer perceptions of kindness, sincerity, friendliness, and helpfulness during service interactions (Peng et al., 2022), which determines customers' trust, purchase intention, loyalty, and word-of-mouth (e.g., Orth et al., 2023; Quach et al., 2022; Tsai and Huang, 2002), and is therefore regarded as a fundamental feature of this industry (Wong and Yang, 2020).

Considering the high-touch features within various hospitality contexts, frontline employees' interactions with customers play an essential role in how customers evaluate service providers and their service quality (Rafiq and Ahmed, 1998). In particular, close interactions with service staff and the extent to which they are friendly and caring during the service process can significantly affect customers' perceived warmth and subsequent reactions (Meuter et al., 2000; Wong and Yang, 2020).

2.2. Change in service providers from human staff to AI facilities

With the development and deployment of new technologies, facilities such as autonomous checkouts, face recognition devices, and service robots have allowed firms to roll out AI-based staff to provide services independently or work alongside frontline employees (Lu et al., 2019). Because machines do not require training or shifts, AI facilities help improve service efficiency, lower operating costs, and enhance service performance (Grewal et al., 2020). Service providers are increasingly deploying AI facilities to replace or facilitate human services, and AI service is gradually becoming more common in the hospitality industry (Jiang and Wen, 2020).

The deployment of AI facilities such as chatbots and robots in the service sector has garnered significant attention from scholars and practitioners (Cheng et al., 2024; Tussyadiah, 2020). Considering the abovementioned benefits of AI facilities, researchers are committed to identifying factors that affect customer acceptance and use of AI applications. Based on the technology acceptance model TAM; Davis et al., (1989), one stream of research has focused on technical features that can improve consumers' perceived usefulness and ease of use, including machine learning abilities and face and voice recognition functions (Huang and Rust, 2018). Another stream of research has revealed that features such as anthropomorphism (Patrizi et al., 2024) and cuteness (Lv et al., 2021) can enhance users' emotional responses to and psychological connections with technologies, which eventually enhances their acceptance of and satisfaction with AI facilities.

Given the critical role of customer emotions and perceived warmth in service encounters (Choi et al., 2021; Chuah and Yu, 2021; Lee and Li, 2023), a growing research stream is dedicated to enhancing perceived warmth and fostering the acceptance and use of AI facilities among customers (e.g., Cheng et al., 2024; El Hedhli et al., 2023; Pham et al., 2024). The notion that perceived service warmth is inherently lower in AI-based services than in human-based services appears intuitive (Lv et al., 2024; Tung and Law, 2017; Yu and Ngan, 2019). While several previous studies have compared consumer attitudes towards these two types of service providers, such as preference (Granulo et al., 2021; Hou et al., 2021) and trust (Patrizi et al., 2024; Pham et al., 2024), direct evidence regarding reduced service warmth and its underlying mechanisms remains elusive. However, service warmth is a defining characteristic of the hospitality industry (Gao and Mattila, 2014) and contributes to overall customer satisfaction (Mittal and Lassar, 1996). Therefore, further examinations of the impacts of AI adoption in this context would be valuable. Accordingly, this research aims to address the following research question: Why and under what circumstances does the transition to AI service compromise customer warmth perception in the hospitality industry?

3. Hypothesis development

3.1. Service providers and perceived warmth

Perceived warmth emerges from interpersonal connections, empathy, understanding, cooperation, caring for others, and ethical behavior between individuals during interactions (Fiske et al., 2002, 2007). Given the high-touch features of hospitality interactions, customers' perceptions of emotional closeness and the extent of their contact with frontline employees dominate their evaluations of service warmth (Güntürkün et al., 2020; Marinova et al., 2018).

In traditional human service contexts, employees naturally embody humanity, displaying emotional responsiveness and interpersonal warmth (Haslam, 2006). By leveraging these innate qualities, employees convey sentiments of love, kindness, attentiveness, and helpfulness during service encounters, thereby fostering perceptions of interpersonal connections, empathy, and understanding (Cheng et al., 2024; Mittal and Lassar, 1996). Consequently, customers experience heightened service warmth.

However, in AI service contexts, interactions are mediated by AI entities that assume roles typically filled by human employees. Although AI entities, such as robots, can deliver high-quality services, their inherent non-human nature limits their capacity to exhibit nuanced emotions, as humans do (Davenport et al., 2020). Consequently, customers engaging with AI (vs. human) service providers are less likely to experience the same level of intimacy and close interactions characterized by human services (Cheng et al., 2024; de Kervenoael et al., 2020; Lv et al., 2024; Wirtz et al., 2018), resulting lower perceived warmth. Thus, we propose the following hypothesis.

H1. AI (vs. human) services negatively affect customer perceived service warmth.

3.2. Mediating effect of the mind perception of the service provider

Although AI demonstrates high autonomy in most service tasks, it remains an artificial construct that lacks the full range of human capabilities (Huang and Rust, 2018; Longoni and Cian, 2022), which may be the key driver of reduced warmth perceptions. Based on the theory of mind perception (Gray et al., 2007; Gray and Wegner, 2012), AI entities are differentiated from human staff based on two components: experience and agency. Specifically, *experience* refers to the capability to feel and sense, while *agency* pertains to the ability to think and intend (Gray et al., 2007; Gray and Wegner, 2012).

Both human and non-human entities can exhibit these features; however, they are primarily regarded as human characteristics (Gray and Wegner, 2012). Consequently, compared to human adults with complete theory of mind, AI facilities are typically perceived as low in experience (Li et al., 2022; Liu and Lee, 2024). For example, people are less likely to accept hedonic products recommended by AI (vs. human) because AI is considered algorithm-based and, therefore, lacks the emotional experiences associated with hedonic products (Longoni and Cian, 2022).

Similarly, AI entities are perceived as low in agency, with many difficulties in identifying and conveying nuanced emotions during interpersonal communication (Morera et al., 2018), leading to negative emotions such as fear and terror (Gray and Wegner, 2012). In addition, even if AI facilities (e.g., service robots) are designed with benevolent intentions, their ability to express such intentions is limited. For example, consumers attribute less credit to virtual influencers than to human influencers predominantly because they believe that AI entities are less responsible for their activities (Liu and Lee, 2024). Similarly, people are more likely to accept an unfavorable offer from an AI agent than from a human agent because less malicious intent is involved in such offers when made by AI rather than humans (Garvey et al., 2023).

Thus, the diminished mind perception of AI service providers, characterized by lower levels of *experience* and *agency*, can result in lower anticipation of warm service interactions compared to human providers. This serves as an essential mechanism underlying diminished perceptions of warmth in AI service encounters. Thus, we propose the following hypothesis.

H2. The mind perception of the service provider mediates the effect of AI (vs. human) services on reduced consumer perceived service warmth.

We hypothesize that replacing human employees with AI staff in the hospitality industry carries a significant risk in reducing service warmth perceptions. Such a negative effect can be attributed to the lower mind perceptions of AI entities as service providers compared to their human counterparts. Therefore, for firms seeking to implement AI facilities (e.g., for operational efficiency) while mitigating potential losses in perceived service warmth, strategies should be considered to elevate customers' warmth perceptions of AI staff or minimize their emphasis on warmth perceptions when evaluating services provided. In the following sections, we propose one boundary condition that could mitigate the negative impact of AI (vs. human) service providers on customer perceptions of service warmth, as well as how to reduce the warmth loss effect on a significant downstream consequence: continuous usage intention.

3.3. Moderating effect of language style

Communication plays a significant role in real-time service encounters and can manifest in either formal or informal styles (Gretry et al., 2017). In formal communication, traditional and standardized languages are employed to convey messages in a more structured manner using formal vocabulary (Oikarinen and Saraniemi, 2016). Employees often adhere to formal communication styles to maintain service quality standards (Han et al., 2015; Sparks and Callan, 1992). Consequently, consumers perceive employees as utilizing formal language and as well trained and deliberate in their service delivery (Barcelos et al., 2018), resulting in heightened perceptions of service warmth.

In contrast, informal communication is characterized as 'common, nonofficial, familiar, casual, and often colloquial' (McArthur, 1992), typically observed in close relationships (Kim et al., 2013). In hospitality contexts, where customers frequently encounter service staff for the first time, the use of informal language is uncommon and can be perceived as lacking thoughtfulness and mindfulness (Choi et al., 2019; Wu et al., 2017), thus diminishing customers' warmth perception.

However, these dynamics may change in AI service settings. Since services delivered by AI facilities are typically viewed as standardized and predictable (Hofmann et al., 2020), it is expected that AI entities will adhere to a formal communication style, potentially delivering a mechanical and non-human impression that reduces customers' warmth perception. However, an informal communication style may serve as a social cue facilitating AI interactions, making consumer–technology exchanges more personal and natural (Feine et al., 2019). Specifically, customers may attribute a higher level of agency and intention to AI entities that communicate in a more informal and human-like manner, resulting in a higher level of service warmth. Therefore, we propose the following hypothesis.

H3. Language style moderates the effect of AI (vs. human) services on perceived service warmth, with the effect attenuated when the AI communicates using an informal (vs. formal) language style.

3.4. Downstream consequence of reduced warmth perception on customers' continuous usage intention

After discussing whether and how AI (vs. human) services reduce customers' perceived warmth, a critical subsequent question emerges: How does this perception influence customers' behavior and attitudes, especially their long-term engagement with these technologies? Given that the integration of AI in hospitality is driven by its long-term operational advantages, we examine a key downstream consequence of reduced warmth perception: lower continuous usage intentions.

Continuous usage intention, distinct from short-term willingness, represents a customers' long-term willingness to use a product or service (Chou et al., 2010). In the AI service context, this refers to a customer's intention to maintain a long-term relationship with the AI staff and the service firm (). As a fundamental dimension for interpersonal evaluation, perceived warmth signifies social acceptance and connection, influencing the tendency to initiate and maintain relationships (Fiske et al., 2007; Lemmink and Mattsson, 2002). Therefore, a decline in perceived warmth negatively impacts relationship maintenance, a dynamic can be extended to human-AI interactions. Existing research demonstrates that reduced warmth perception from AI weakens customers' intention to continue using it (Hsieh and Lee, 2024; Zhou et al., 2023). Consequently, when service providers transit from human to AI staff, the reduced perceived warmth, stemming from lower mind perception of the AI service provider, is expected to diminish continuous usage intention. Taken together, we propose the following hypothesis.

H4. The mind perception of the service provider and perceived service warmth mediate the effect of AI (vs. human) services on reduced customer's continuous usage intention.

3.5. Moderating effect of customer motivation

Customer motivations in different consumption contexts can be classified as hedonic or utilitarian (Dhar and Wertenbroch, 2000; Hartman et al., 2006). Specifically, a *utilitarian motivation* is task-oriented and inspired by functional needs, whereas a *hedonic*

motivation is inspired by the desire for fun and sensual pleasure (Hirschman and Holbrook, 1982; O'Curry and Strahilevitz, 2001). Thus, utilitarian-oriented customers seek functionality and efficiency, whereas hedonic-oriented customers seek social interaction and entertainment (Yim et al., 2014). Both motivations can coexist in most cases, although the predominant motivation varies according to the characteristics of a situation.

Customers' motivational orientation determines their preferences and behaviors across various contexts. For example, Büttner et al. (2015) found that hedonic-oriented customers are more attracted to non-monetary promotions than utilitarian-oriented customers. More specifically, Chang et al. (2023) suggested that both hedonic and utilitarian motivations may influence customers' purchase intentions in smart stores. Specifically, utilitarian motivation is more formative, leading customers attention to product features such as price, quality, location convenience, shopping speed, and product recommendations. In contrast, hedonic motivation is more reflective, emphasizing experiences of control, curiosity, joy, focused immersion, and temporal dissociation.

Therefore, the impact of perceived service warmth on customers' continuous usage intention is likely contingent on their motivational orientation. For customers with hedonic motivations, who prioritize the emotional aspects of service encounters, reduced warmth perception is expected to significantly influence their core values and, consequently, their intentions to continue using the services. In contrast, utilitarian-oriented customers, focusing on service efficiency and outcomes rather than on sensory experiences. As a result, a decrease in perceived service warmth is less likely to affect their core values or continuous usage intentions. To sum up, we propose the following hypothesis.

H5. Customer motivation moderates the indirect effect of AI (vs. human) service on customers' continuous usage intention through perceived service warmth, with the effect is attenuated for customers with a utilitarian (vs. hedonic) motivation.

Fig. 1 demonstrates the conceptual framework of our research.

4. Overview of the studies

To test our hypotheses, we conducted five studies (and one supplemental study), as summarized in Table 1. Based on real online customer reviews of either human or AI service hotels, Study 1 provided empirical evidence that customers perceived lower service warmth when staying at AI service hotels than when staying at human service hotels. The study reported in the Supplementary further confirmed this relationship using a field survey. Using a scenario-based experiment, Study 2 demonstrated the causal impact of AI (vs. human) services on lower perceived service warmth across both Chinese and US participants. Study 3 identified that the mind perception of the service provider is the underlying mechanism of the AI service effect. Building on these findings, Study 4 suggested a mitigation strategy: the adverse warmth loss effect was attenuated when AI (vs. human) staff communicated in an informal (vs. formal) language style. Finally, Study 5 extended the investigation by demonstrating reduced warmth perceptions in AI service encounters lead to lower continuous usage intentions. However, this negative downstream effect was mitigated for consumers with a utilitarian (vs. hedonic) motivation.

Table 1
Overview of studies.

Study	Objective	Method	Location in paper
Study 1	Main effect, H1	Text analysis of online customer reviews	Main text
Supplemental study	Main effect, H1	Field survey	Web appendix B and Discussion of Study 1
Study 2	Main effect, H1	Scenario-based experiment	Main text
Study 3	Mediating effect, H2	Scenario-based experiment	Main text
Study 4	Moderating effect, H3	Scenario-based experiment	Main text
Study 5	Downstream consequence and moderating effect, H4, H5	Scenario-based experiment	Main text

4.1. Study 1: online customer reviews and text analysis

Customers nowadays tend to leave reviews online to share their authentic experiences and feelings when staying at hotels. This is also a practice for customers to give feedback to hotel managers. While these comments were not explicitly generated to evaluate consumers' perceived warmth, they still convey it. More importantly, these comments were written by a vast range of customers who stayed at different hotels, reflecting the general changes in industrial warmth. Prior research has verified that online customer reviews have good reliability and validity (Edwards et al., 2018). Therefore, Study 1 aimed to test H1 by analyzing customer reviews posted on Ctrip (www.ctrip.com), a well-known online booking platform in China, to preliminarily examine the impact of service provider transformation on customers' perceived warmth.

4.1.1. Method

We first searched keywords such as "AI," "robot," "intelligent," "smart," and "humanless" on Ctrip to identify AI-service hotels in first-tier cities in China (i.e., Beijing, Shanghai, Guangzhou, and Shenzhen). We then manually evaluated whether the hotel is an AI-service hotel through its introduction, displayed pictures, and customer reviews. Finally, we identified 63 hotels that provided *real* AI service. Then, we identified regular hotels in the same business district as each AI hotel (within 3 km) and selected the one with the most similar price ranges and customer ratings. In total, we identified 63 human-service hotels as well. The prices of these hotels ranged from 100 RMB to 1000 RMB, including both high-end and budget hotels.

A total of 33,342 reviews of AI-service hotels and 38,592 reviews of human-service hotels were captured using Python on March 4, 2021. We

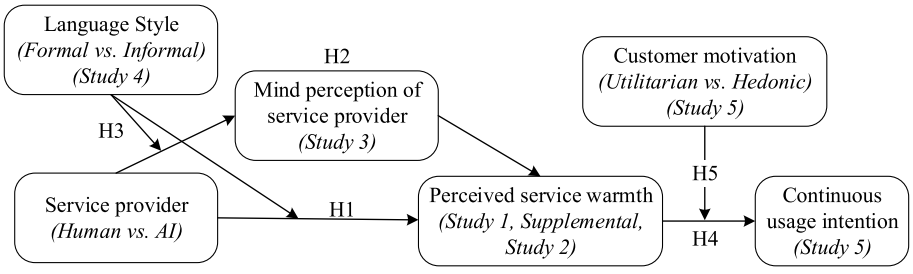


Fig. 1. Conceptual framework
Fig. 1 Alt Text: Conceptual framework showing the relationships between variables, the five hypotheses, and the studies designed to test each hypothesis.

then used content analysis to rate customers' perceived warmth of the selected 126 hotels. To establish a proper rating policy for perceived warmth, we captured another 30,000 customer reviews on the 200 most popular hotels in Beijing. All customer reviews were posted within one year based on the check-in dates. Thirty graduate students majoring in Hospitality and Service Management were invited to extract the warmth-related vocabulary. Training sessions were conducted to ensure each student had a good understanding of warmth perception based on various measurements and definitions of warmth (e.g., warm, kind, friendly, and sincere) in prior research (e.g., Fiske et al., 2007; Wang et al., 2017). Whereafter, each student first evaluated 1000 comments randomly selected from the pool, resulting in a total of 318 warmth-related words. Then, four experts in the same field were asked to classify these words into two groups: warm (e.g., warm-hearted, feeling at home, super warm) and cold (e.g., cold-hearted, indifferent, fearful), and rated the intensity of each word from low to high (warm: 1–10, 1 = slightly warm, 10 = very warm; cold: −10 ~ −1, −1 = slightly cold, −10 = very cold). Results of the Kendall's coefficient of concordance verified a high agreement among all the raters (Kendall's w = .884, $p < .001$). Finally, a vocabulary with 237 warm-related words (ranging from 1 to 10) and 81 cold-related words (ranging from −1 ~ −10) was obtained.

We use Python to identify the warmth-related words, calculate the perceived warmth score of each customer review, and form the average perceived warmth for each of the 126 hotels based on the obtained vocabulary. Specifically, this study employed the open-source Python package (*Jieba*, https://github.com/fxsjy/jieba) to perform pre-processing operations such as word splitting, lexical annotation, and removal of deactivated words on hotel review texts. *Jieba* is a widely used third-party Chinese word separation dictionary in Python, recognized for its effectiveness in Chinese text mining and sentiment analysis (e.g., Zhao et al., 2023). The warmth score of each review was the sum of the scores of all warmth- (or cold-) related words in the review. The warmth score for each hotel was obtained by dividing the warmth score of all reviews by the number of reviews. In Web Appendix A, we presented ten sample reviews and the perceived warmth score of each review.

4.1.2. Results and discussion

We first matched each AI-service hotel with a human-service hotel based on distance and price level and then conducted a paired-sample t-test. The results showed that the average customer perceived warmth of the AI-service hotels was significantly lower than that of the human-service hotels (M_{AI} = 6.88 vs. M_{human} = 8.37; t = −2.94, $p < .01$, Cohen's d = .37). Thus, H1 is supported.

Using online customer reviews and text analysis, Study 1 provided real-life evidence that AI (vs. human) service reduces customer perceived warmth in hotels varied in price levels and brand images. It also reflected the ubiquity of this problem in the hospitality industry, considering the large size and diverse sources of online customer reviews under examination.

To further confirm our predictions, we conducted a field survey (N = 94; 45.7% female, M_{age} = 33.45) outside an AI-service hotel and a human-service hotel in a large city in China (i.e., Hangzhou). The two hotels were similar in price range and located in the same business circle. Customers of the two hotels were invited to participate in this study via the convenience sampling method and were approached after checking out. They were asked to indicate their perceived service warmth when staying at the specific hotel and reported their demographics. As expected, results of one-way ANOVA showed that customers staying at the AI-service hotel reported significantly lower perceived warmth than those staying at the regular, human-service hotel (M_{AI} = 3.91 vs. M_{human} = 5.39; $F(1, 92) = 28.05$, $p < .001$, $\eta^2 = .23$) (see Web Appendix B for more details).

Together, these two studies provide preliminary evidence that AI (vs. human) service reduces customers' perceived warmth. However, one may argue that customers self-selected into either an AI-service hotel or a human-service hotel, and factors such as individual traits and hotel brand image might drive the observed effect. We address this issue in Study 2 using a scenario-based experiment.

4.2. Study 2: The main effect of AI service on perceived warmth

Study 2 aimed to provide causal evidence that AI service reduces service warmth, while addressing concerns about hotel brand image and individual differences. To this end, we created two videos of a hotel's service processes, either delivered by a human staff or a robot. Both videos were produced at Ascott Raffles City Chengdu Hotel to control confounding factors such as the hotel environment. In addition, to validate the generalizability of our findings across different cultures, Study 2 collected data from both China and the United States.

4.2.1. Participants and procedures

Study 2 adopted a one-factor (service provider: human vs. AI) between-subjects design. Since this was our first experimental study, we conducted a statistical power analysis using G*Power Program 3.1 to estimate the sample size for a one-factor between-subjects design (F-test [ANOVA: Fixed effects, omnibus, one-way]). Results revealed that a minimum sample size of 90 was required for a medium effect size of .30, at significance level of .05, to achieve a statistical power of .80. To this end, a total of 127 Chinese participants (55.1% female, M_{age} = 32.50) recruited from a large online panel (www.wjx.com) and 132 US participants (46.2% female, M_{age} = 34.89) recruited from Mturk, who passed the attention check, completed this study for a monetary reward. Participants were recruited using simple random sampling techniques in both conditions, as were the subsequent studies.

Participants were randomly assigned to either the human or AI service condition and then watched a corresponding video. In the AI-service video, an AI-enabled service robot greeted the customer, guided him/her to the elevator, and delivered a bottle of water to the guest room. In the human-service video, an experimenter was hired as the hotel staff and did what the robot did in the AI condition.[1] After then, participants indicated their perceived service warmth on a 4-item scale (i.e., warm, kind, friendly, sincere; China: α = .81; US: α = .85) adapted from Wang et al. (2017). An attention check question "*Please choose '2' to show that you are reading carefully*" was included as a quality control. Finally, participants provided their demographic details such as age and gender.

4.2.2. Results and discussion

Main effect. One-way ANOVA results verified that participants in the AI service condition reported significantly lower perceived warmth than those in the human service condition, for participants both from China (M_{AI} = 5.46 vs. M_{human} = 5.71; $F(1, 125) = 8.13$, $p < .01$; $\eta^2 = .06$) and the United States (M_{AI} = 5.58 vs. M_{human} = 5.90; $F(1, 130) = 4.81$, $p < .05$, $\eta^2 = .04$). Additional regression analysis on participant's perceived warmth with service provider type (AI vs. human) as independent variable, while controlling for participants' age and gender, further proved the negative effect of AI service (China: $\beta = -.24$, $t = -2.75$, $p < .01$; US: $\beta = -.19$, $t = -2.24$, $p < .05$). Thus, H1 was supported. The results remained significant after controlling for participants' demographics (see Web Appendix C).[2]

Using a controlled scenario experiment, Study 2 provided causal evidence that changing the service provider from human to AI staff (i.e., service robots in particular) led to decreased customer perceived

[1] For the human-service video, please refer to https://www.bilibili.com/video/BV1gM4y1d7wn/; for the AI-service video, please refer to https://www.bilibili.com/video/BV1Bj411g7Rj/.

[2] Including participants' demographical characteristics did not change the pattern of relationships of this study and all subsequent studies.

warmth. Furthermore, we replicated the same pattern of relationship across both Chinese and US sample groups, thereby excluding culture as a potential confounder.

4.3. Study 3: The mediating effect of mind perception of the service provider

Using different methods, namely text analysis, field survey, and scenario-based experiment, prior studies provided both correlational and causal evidence for the negative effect of AI (vs. human) service on perceived warmth, across different cultural groups. In Study 3, we aimed to examine the mediating role of mind perception of the service provider to test H2.

4.3.1. Participants and procedures

Study 3 employed a one-factor (service provider: human vs. AI) between-subjects design. A total of 213 participants (48.8% female, M_{age} = 28.04), who were recruited from an online panel (www.wjx.com) and passed the attention check, completed the study for a monetary reward.

Participants were randomly assigned to either the AI service or the human service condition, and then read a collage of pictures showing a service staff or AI-enabled service robot providing the same services in the hotel, such as greeting customers, catering food, and room service (see Web Appendix D). After that, they indicated their mind perceptions of the service provider (either human staff or service robots) using an 18-item scale (i.e., *Experience*: hunger, fear, pain, pleasure, rage, desire, personality, consciousness, pride, embarrassment, and joy; α = .92; *Agency*: self-control, morality, memory, emotion recognition, planning, communication, and thought; α = .90) from Gray et al. (2007). As we did in Study 2, participants completed the same measurement of perceived service warmth (α = .88). Results of the reliability and validity tests for all the scales, reported in Web Appendix E, verified the choices of these scales. Finally, participants answered the same attention check question and provided their demographics.

4.3.2. Results and discussion

Main effect. The one-way ANOVA results showed that, compared to participants in the human service condition, those in the AI-service scenario reported lower perceived warmth (M_{AI} = 4.59 vs. M_{human} = 5.70; $F(1, 211) = 70.95$, $p < .001$, $\eta^2 = .25$) and lower mind perception of the service provider (*Experience*: M_{AI} = 4.37 vs. M_{human} = 5.41; $F(1, 211) = 83.48$, $p < .001$, $\eta^2 = .28$; *Agency*: M_{AI} = 4.47 vs. M_{human} = 5.50; $F(1, 211) = 71.02$, $p < .001$, $\eta^2 = .25$). Thus, H1 was supported.

Mediation effect. To test H2, we employed the bootstrapping analysis via the PROCESS macro (Model 4; 5000 samples; Hayes, 2013), with service provider type (human vs. AI) as the independent variable, mind perception of the service provider as the mediator, and perceived warmth as the dependent variable. The results revealed a significant indirect effect of the service provider on perceived warmth via mind perception (Experience: *indirect effect* = −.67, 95% *CI* = [−.87, −.50]; Agency: *indirect effect* = −.46, 95% *CI* = [−.61, −.32]), supporting H3. When mind perception was included in the regression, the direct effect of AI (vs. human) service provider was not significant (*direct effect* = −.01, 95% *CI* = [−.15, .12]).

Study 3 provided evidence for the mechanism underlying the AI service effect. Specifically, AI (vs. human) service shed a negative light on customers' perceived warmth because it hampers customers' mind perception of the service provider (i.e., the AI), revealing the essential role of human contact in hotel services.

4.4. Study 4: The moderating role of language style

Study 4 aimed to test H3, which suggests that the negative effect of AI (vs. human) service providers can be mitigated in the era of new technology, by featuring an informal (vs. formal) language style in which the AI staff communicates with customers.

4.4.1. Participants and procedures

Study 4 employed a 2 (service provider: human vs. AI) × 2 (language style: formal vs. informal) between-subjects design. Since this is our first two-factor between-subjects experiment, we therefore conducted a statistical power analysis to estimate the sample size using G*Power 3.1 (F-test [ANOVA: Fixed effects, special, main effects and interactions]; Faul et al., 2009). Results revealed that, for a medium effect size of .30, at a significance level of .05, a minimum sample size of 90 was required to achieve a statistic power of .80. To this end, a total of 450 participants (49.3% female, M_{age} = 30.28), who passed the attention check, were recruited from the same online panel of Study 2.

Participants were first invited to evaluate the service of a hotel and then randomly assigned to one of the four conditions. They watched the corresponding video filmed at the same hotel as Study 2. In the video, a service robot or a service employee greets customers in different language styles.[3] Following Gretry et al. (2017), the informal language style was manipulated using informal vocabulary and informal punctuation (e.g., "We" "!" and "Thanks"), whereas the formal language style was manipulated using normative words and punctuations (e.g., "You" "." and "Thank you"). For example, in the informal language condition, the service staff (AI) greeted the customer and said: "*Hi! We're located in the city center, with easy transportation, many different room types and amenities, and awesome I. Basically, we can meet all your needs. If you need anything, just give us a shout, and we're always here to help*." In contrast, in the formal language condition, the service staff (AI) said: "*Dear customer, our hotel is conveniently located in the city center, with a variety of room types, complete facilities, and exquisite decorations to meet your different needs. If you have any requests during your stay, please feel free to call our hotel's service hotline. All hotel staff (robots) will be dedicated to serving you*."

After watching the video, participants reported their perception of the language style used by the service provider (human or AI) via a 4-item scale (i.e., informal, unofficial, casual, and easy-going; α = .95) adopted from Gretry et al. (2017). Then, using the same scales detailed in Study 3, participants reported their mind perceptions of the service provider (*Experience*: α = .90; *Agency*: α = .80) and perceived warmth (α = .82).[4] Finally, participants answered the attention check question before providing their demographics.

4.4.2. Results and discussion

Manipulation check. The one-way ANOVA results showed that participants in the informal language style condition indicated that the language style was more informal than those in the formal condition ($M_{informal}$ = 5.08 vs. M_{formal} = 3.91; $F(1, 448) = 68.42$, $p < .001$). Therefore, our manipulation was successful.

The main effect. The one-way analysis of variance (ANOVA) results showed that AI service was perceived as less warm than human service (M_{AI} = 5.53, vs. M_{human} = 5.68; $F(1, 448) = 5.93$, $p < .05$). Thus, H1 was supported.

The moderating effect of the language style. Results of the two-way ANOVA on perceived warmth yielded a main effect of the service provider ($F(1, 446) = 4.55$, $p < .05$), but a null effect of language style ($F(1, 446) = 1.06$, $p > .05$). More importantly, the interaction effect was significant ($F(1, 446) = 14.00$, $p < .001$). Specifically, when communicating in a formal language style, AI (vs. human) service resulted in

[3] For the video of human service in a formal language style: https://www.bilibili.com/video/BV1Uo4y1r78i/; for the video of AI service in a formal language style: https://www.bilibili.com/video/BV1E24y137QL/; for the video of human service in an informal language style: https://www.bilibili.com/video/BV1HL41117nv/; for the video of AI service in an informal language style: https://www.bilibili.com/video/BV1LY411k7VR/.

[4] For detailed reliability and validity results, please refer to Web Appendix E.

lower perceived warmth (M_{AI} = 5.37 vs. M_{human} = 5.76; $F(1, 446)$ = 18.96, $p < .001$, $\eta^2 = .08$), replicating the findings of previous studies. Such a difference, however, was not observed when the AI staff adopted an informal language style (M_{AI} = 5.68 vs. M_{human} = 5.59; $F(1, 446)$ = 1.22, $p > .05$, $\eta^2 < .01$) (see Fig. 2). Thus, H3 was supported.

The moderated mediation effect. Next, we employed the bootstrapping analysis via the PROCESS macro for SPSS (Model 8; 5000 samples; Hayes, 2013), with the service provider as the independent variable, language style as the moderator, mind perception of the service provider as the mediator, and perceived warmth as the dependent variable. Results revealed a significant moderated moderation effect (Experience: *index* = .15, 95% *CI* = [.04, .30]; Agency: *index* = .12, 95% *CI* = [.02, .26]). Specifically, in the formal language style condition, the mediating effect of mind perception (Experience: *indirect effect* = −.14, 95% *CI* = [−.22, −.07]; Agency: *indirect effect* = −.14, 95% *CI* = [−.26, −.05]) was significant. In the informal language style condition, however, the mediating effect was not significant (Experience: *indirect effect* = .02, 95% *CI* = [−.03, .07]; Agency: *indirect effect* = .01, 95% *CI* = [−.07, .09]). Thus, H4 was supported.

Study 4 found the negative effect of AI service on customers' perceived warmth can be mitigated when the AI staff communicates with customers in an informal, rather than formal, language style. This is because informal language could be a social cue that makes the customer-AI interactions more personal and natural, leading to higher levels of mind perception and perceived warmth.

4.5. Study 5: The Moderating effect of customer Motivation

The main objectives of Study 5 are twofold. First, we aim to reveal a significant downstream effect of reduced warmth perceptions in AI service encounters, namely lower continuous usage intentions, thereby testing H4. In addition, we propose to show such an effect will be mitigated for consumers with a utilitarian (vs. hedonic) motivation, and thus providing support for H5.

4.5.1. Pretest

To verify the effectiveness of the customer motivation manipulation, we conducted a pretest (N = 60, 53.3% female, M_{age} = 25.58) from the same population before Study 5. Following prior research (Lv et al., 2020), participants were told that they booked a hotel for business or leisure purposes and then indicated the extent to which they expected the hotel service to be utilitarian (i.e., effective, helpful, functional, necessary, practical; $\alpha = .90$) or hedonic (i.e., fun, exciting, delightful, thrilling, enjoyable; $\alpha = .91$), on a 7-point Likert scale with 1 = *not at all* and 7 = *very much* (Voss et al., 2003). The paired sample *t*-test results showed that participants in the business trip condition expressed stronger utilitarian motivation than hedonic motivation ($M_{utilitarian}$ = 4.94 vs. $M_{hedonic}$ = 4.59; $t = 2.53$, $p < .05$), while those in the leisure trip condition expressed higher hedonic motivation than utilitarian motivation ($M_{hedonic}$ = 5.24 vs. $M_{utilitarian}$ = 4.07; $t = 6.57$, $p < .001$). In addition, compared to their business-travel counterparts, the leisure-travel participants expressed higher hedonic motivation ($M_{business}$ = 4.59 vs. $M_{leisure}$ = 5.24; $t = 2.28$, $p < .05$) but lower utilitarian motivation ($M_{business}$ = 4.94 vs. $M_{leisure}$ = 4.07; $t = 3.02$, $p < .01$). Thus, the customer motivation manipulation worked as expected.

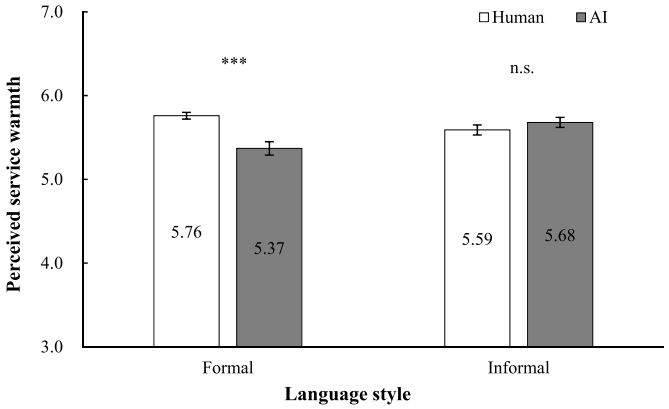


Fig. 2. The interaction effect of service provider and language style conditions on customers' perceived service warmth (Study 4)
Note: ***$p < .001$, n.s. = not significant; the error bar represents the standard error (SE).
Fig. 2 Alt Text: The differences in customers' perceived warmth across service provider (AI vs. human) and language style (formal vs. informal) conditions.

4.5.2. Participants and procedures

Study 5 employed a 2 (service provider: human vs. AI) × 2 (customer motivation: utilitarian vs. hedonic) between-subjects design. A total of 232 participants (51.3% female, M_{age} = 28.19), recruited from the same online panel of Study 2, were randomly assigned to one of the four conditions.

As verified in the pretest, we manipulated customer motivation by informing participants their purpose of staying at the hotel was either for a business trip (utilitarian motivation condition) or a leisure trip (hedonic motivation condition). Participants were presented with pictures of a hotel providing either human service or AI service, as used in Study 3. Next, participants reported their continuous usage intention of the service via a 3-item scale ("I will try the human/AI service again, not just abandon it," "I will continue to use the human/AI service, not calling for anyone's help," and "If possible, I will stop using this human/AI service"; α = .77) adopted from Bhattacherjee (2001). As in previous studies, participants also indicated their mind perception of the (human or AI) service provider (*Experience*: α = .89; *Agency*: α = .78) and perceived warmth (α = .79). Finally, participants reported their demographic details such as age and gender.

4.5.3. Results and discussion

The main effect. Results of one-way ANOVA showed that AI service was perceived as less warm than human service (M_{AI} = 4.47, vs. M_{human} = 5.62; $F(1, 230)$ = 179.86, $p < .001$), replicating the findings of previous studies.

The downstream effect on customers' continuous usage intention. The follow-up analysis further supported that AI service reduced customers' continuous usage intention (M_{AI} = 4.24, vs. M_{human} = 5.56; $F(1, 230)$ = 179.36, $p < .001$). We then employed the bootstrapping analysis (PROCESS Model 80; 5000 samples; Hayes, 2013), with service provider type (human vs. AI) as independent variable, mind perception (*experience* and *agency*) and perceived warmth as serial mediators, and continuous usage intention as dependent variable. Results confirmed that AI (vs. human) service resulted in lower reusage intentions due to lower mind perceptions and warmth perceptions of AI as service providers (Experience and warmth: *indirect effect* = −.44, 95% *CI* = [−.61, −.30]; Agency and warmth: *indirect effect* = −.23, 95% *CI* = [−.35, −.13]). Thus, H4 was supported.

The moderating effect of customer motivation. To test the moderating role of customer motivation, we first conducted a two-way ANOVA on participants' continuous usage intention of AI or human service. Results yielded a main effect of service provider ($F(1, 228)$ = 192.49, $p < .001$), but a null effect of customer motivation ($F(1, 228)$ = 2.02, p = .16). More importantly, the interaction effect was significant ($F(1, 228)$ = 14.92, $p < .001$). Specifically, for participants with a hedonic motivation (i.e., in a leisure trip), they indicated lower continuous usage intention in the AI service condition than those in the human service condition (M_{AI} = 3.98 vs. M_{human} = 5.67; $F(1, 113)$ = 118.02, $p < .001$, $\eta^2 = .51$). Such a difference, though remained significant, was largely reduced for participants with a utilitarian motivation in a business trip (M_{AI} = 4.49 vs. M_{human} = 5.44; $F(1, 115)$ = 73.86, $p < .001$, $\eta^2 = .39$) (see Fig. 3).

The moderated mediation effect. To further examine the moderated

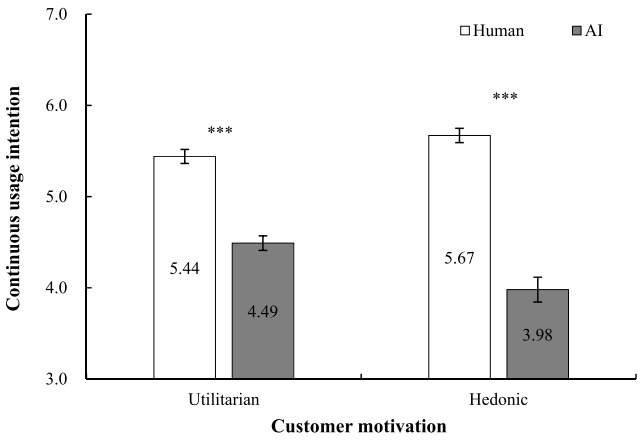


Fig. 3. The interaction effect of service provider and customer motivation conditions on customers' continuous usage intention (Study 5)
Note: ***$p < .001$, **$p < .01$; the error bar represents the standard error (SE).
Fig. 3 Alt Text: The differences in customers' continuous usage intention across service provider (AI vs. human) and customer motivation (utilitarian vs. hedonic) conditions.

mediation effect, we employed the bootstrapping analysis (PROCESS Model 14; 5000 samples; Hayes, 2013), with service provider (AI vs. human) as the independent variable, perceived warmth as the mediator, customer motivation as the moderator, and continuous usage intention as the dependent variable. As expected, results revealed a significant moderated mediation effect (*index* = −.26, 95% *CI* = [−.49, −.06]). Specifically, in the hedonic condition, the mediating effect of perceived warmth (*indirect effect* = −.89, 95% *CI* = [−1.14, −.67]) was significant. In the utilitarian condition, however, the mediating effect was largely reduced (*indirect effect* = −.63, 95% *CI* = [−.82, −.45]), though remained statistically significant (see Fig. 3). Thus, H5 was supported. We reported the results of bootstrapping analysis taking mind perception and perceived service warmth as serial mediators in Web Appendix F.

Study 5 proved again that a transition from human service to AI service resulted in lower perceived warmth, which showed a negative downstream impact on customers' intention to continue using AI service. However, such an effect varied across customers with different motivations. Specifically, compared to hedonic-oriented customers, the negative effect of perceived warmth on reduced reusage intention was mitigated for utilitarian-oriented customers. This is because customers with a utilitarian motivation emphasize service efficiency and outcome more over emotional experiences and human contact and, thus are less likely to be affected by the AI transformation.

5. Discussion and conclusions

5.1. Conclusions

Perceived warmth is a core element distinguishing the hospitality sector from others (Peng et al., 2022). However, with the growing deployment of AI facilities such as chatbots and service robots, interactions between customers and service employees, which are recognized as the primary sources of perceived service warmth, have gradually been replaced by customer–technology encounters (Huang and Rust, 2018; Lee and Li, 2023). With customers increasingly accessing services from AI facilities while contacting fewer human employees in the hospitality industry (Jiang and Wen, 2020), such an AI transition might have unintended negative consequences.

Using text analysis, a field survey (in the supplementary), and four scenario-based experiments, this research provides robust evidence for reduced warmth perceptions as a consequence of AI deployment. The results of Study 1 (and the supplemental study) suggested that, compared with customers who stayed at human-service hotels, those staying at AI-service hotels experienced significantly lower perceived warmth. Study 2 confirmed the causal relationship between AI (vs. human) services and reduced service warmth across different cultures (i.e., China and the US). Study 3 demonstrated that the lower-level mind perceptions (both agency and experience) of AI (vs. human) as a service provider underlie the reduced perceived warmth effect. Finally, the results of Studies 4 and 5 suggested two strategies for service firms that seek AI deployment while avoiding the loss of warmth perceptions: (1) installing AI facilities that communicate with customers in an informal (vs. formal) language style, and (2) targeting customers with a utilitarian (vs. hedonic) motivation.

5.2. Theoretical contributions

First, our study contributes to the existing literature that has primarily emphasized the benefits of AI deployment by revealing an adverse effect on customers' warmth perception. In the hospitality industry, rational judgments (e.g., of service efficiency) and emotional judgments (e.g., of perceived warmth) are important predictors of customer evaluations, with emotional elements playing a more critical role (Güntürkün et al., 2020). Several prior studies have revealed that AI service may lead to negative consequences such as lower satisfaction, reduced loyalty, and lower word-of-mouth intention (e.g., Longoni et al., 2019; Moore et al., 2022; Prentice and Nguyen, 2020). By focusing on perceived service warmth, a fundamental element of the hospitality industry, our study reveals reduced service warmth as a new negative effect of AI deployment, and lower continuous usage intention as a downstream consequence.

Second, by revealing the role of mind perceptions in the negative effect of AI service on customer perceived warmth, this study further underscores the importance and uniqueness of human contact in the hospitality industry. Specifically, we find that the key reason for reduced service warmth after the AI transition is because customers tend to believe that AI service providers (e.g., robots) have lower levels of mind perceptions (both experience and agency) than human staff. We have thus disclosed a new psychological mechanism behind the reduced service warmth caused by the human–AI transition and deepened the current understanding of this phenomenon. Our research provides theoretical guidance for improving perceived warmth in the new technology era and for mitigating, if not fully solving, problems.

Third, our work contributes to the literature on customer–technology interactions by identifying the moderating roles of language style and customer motivation in service encounters. Although previous studies suggest that customers prefer a formal language style in the service context (Han et al., 2015; Sparks and Callan, 1992), our research suggests that an informal communication style, rather than a formal style, enhances customers' mind perceptions of AI staff, which consequently mitigates the warmth loss effect in AI service encounters. In addition, we further reveal that a utilitarian (vs. hedonic) motivation will reduce customers' focus on emotional connections and human contact in service encounters. Consequently, reduced service warmth is less likely to negatively influence customers' continuous usage intentions of AI service.

5.3. Practical implications

This study also provides valuable insights for hotel managers regarding AI deployment. First, despite the benefits of service efficiency and reduced operational costs, hotel managers should be aware of the negative effects of the transition from human to AI service on customers' feelings and warmth perceptions. AI service can significantly reduce human contact, which plays a vital role in customer warmth perceptions. Therefore, in the process of automation upgrading, hotel managers should be more cautious and keep human staff in significant

positions to maintain a certain level of interpersonal interaction and customer perceived service warmth.

Second, hotel managers should ascertain the mind perception that customers require from service providers to ensure customer satisfaction. In many cases, customers may value human staff who provide flexible and friendly services. However, in certain service contexts requiring higher levels of security and convenience, customers may prefer AI service despite lower mind perception, such as information provision, housekeeping activities, and booking (Ivanov and Webster, 2019). Hence, hotel managers can strategically assign their human staff and AI facilities to different service tasks based on customer perceptions and requirements to guarantee efficient service delivery and perceived service warmth.

Third, service firms aiming to leverage the benefits of AI deployment (e.g., for lower operation costs and higher efficiency) while avoiding the potential risks of warmth loss could consider two practical strategies. First, hotel managers could collaborate with technology developers to customize the language style (e.g., using more informal vocabulary and punctuation, emoticon lexical bundles, and sound mimicking) of AI facilities to improve customers' mind perception and diminish the negative effect of AI service on customer perceived warmth. Additionally, service firms deploying AI service can provide more utilitarian (vs. hedonic) services or target utilitarian-oriented (vs. hedonic-oriented) customers to reduce their emphasis on emotional experiences and human contact in service encounters, thus mitigating the adverse effects of reduced warmth perceptions on customers' continuous usage intentions.

5.4. Limitations and future research directions

This current research entails several limitations. First, despite various sectors in the hospitality industry (e.g., hotels, restaurants, and tourism attractions), we examined only the effects of AI service on customer perceived warmth in hotels. However, customer–technology interactions may exert different effects across various service contexts, and whether the negative impact on warmth perception in the hotel industry will be observed in other hospitality contexts requires further investigation. Second, in most service contexts, AI facilities, such as service robots, work alongside human employees. Therefore, situations such as the level of AI replacement (e.g., full vs. partial) and roles of AI facilities (e.g., coworker vs. assistant vs. manager) need further explorations. Third, in the current research (especially Studies 2–5), we mainly focused on service robots as AI facilities deployed in hotels and found that robot adoption was associated with a reduction in customer perceived warmth. Future research can explore whether other AI facilities, such as chatbots and avatars that exhibit more human-like features (e.g., face and voice) and nuanced emotional expressions, can address the problem of warmth loss observed in the current research.

CRediT authorship contribution statement

Xingyang Lv: Writing – original draft, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Yufan Yang:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation. **Dazhi Qin:** Writing – original draft, Conceptualization. **Xiaoyan Liu:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Conceptualization.

Declaration of competing interest

We wish to confirm that there are no known conflicts of interest associated with this manuscript and there has been no significant financial support for this work that could have influenced its outcome.

Acknowledgement

This work was supported by [Humanities and Social Science Fund of Ministry of Education of China No. 23YJC790088] and [Fundamental Research Funds for the Central Universities of China No. JBK2406039] under Grant [Guanghua Talent Project] to Xingyang Lv; the [National Natural Science Foundation of China] under Grant [No. 72302191] to Xiaoyan Liu.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104282.

Data availability

Data will be made available on request.

References

Aaker, D.A., Stayman, D.M., Hagerty, M.R., 1986. Warmth in advertising: measurement, impact, and sequence effects. J. Consum. Res. 12 (4), 365–381.
Barcelos, R.H., Dantas, D.C., Sénécal, S., 2018. Watch your tone: how a brand's tone of voice on social media influences consumer responses. J. Interact. Market. 41 (1), 60–80.
Bhattacherjee, A., 2001. Understanding information systems continuance: an expectation-confirmation model. MIS Q. 25 (3), 351–370.
Bilal, M., Zhang, Y., Cai, S., Akram, U., Halibas, A., 2024. Artificial intelligence is the magic wand making customer-centric a reality! An investigation into the relationship between consumer purchase intention and consumer engagement through affective attachment. J. Retailing Consum. Serv. 77, 103674.
Buhalis, D., Sinarta, Y., 2019. Real-time co-creation and nowness service: lessons from tourism and hospitality. J. Trav. Tourism Market. 36 (5), 563–582.
Büttner, O., Florack, A., Goeritz, A., 2015. How shopping orientation influences the effectiveness of monetary and nonmonetary promotions. Eur. J. Market. 49 (1–2), 170–189.
Chang, Y.-W., Hsu, P.-Y., Chen, J., Shiau, W.-L., Xu, N., 2023. Utilitarian and/or hedonic shopping–consumer motivation to purchase in smart stores. Ind. Manag. Data Syst. 123 (3), 821–842.
Cheng, Z., Fan, W., Shao, B., Jia, W., Zhang, Y., 2024. The impact of intelligent customer service agents' initial response on consumers' continuous interaction intention. J. Retailing Consum. Serv. 76, 103585.
Chi, O.H., Denton, G., Gursoy, D., 2020. Artificially intelligent device use in service delivery: a systematic review, synthesis, and research agenda. J. Hospit. Market. Manag. 29 (7), 757–786.
Choi, S., Liu, S.Q., Mattila, A.S., 2019. 'How may I help you?' Says a robot: examining language styles in the service encounter. Int. J. Hospit. Manag. 82, 32–38.
Choi, S., Mattila, A.S., Bolton, L.E., 2021. To err is human (-oid): how do consumers react to robot service failure and recovery? J. Serv. Res. 24 (3), 354–371.
Chou, S.W., Min, H.T., Chang, Y.C., Lin, C.T., 2010. Understanding continuance intention of knowledge creation using extended expectation–confirmation theory: an empirical study of Taiwan and China online communities. Behav. Inf. Technol. 29 (6), 557–570.
Chuah, S.H.W., Yu, J., 2021. The future of service: the power of emotion in human-robot interaction. J. Retailing Consum. Serv. 61, 102551.
Davenport, T., Guha, A., Grewal, D., Bressgott, T., 2020. How artificial intelligence will change the future of marketing. J. Acad. Market. Sci. 48, 24–42.
Davis, F.D., Bagozzi, R.P., Warshaw, P.R., 1989. User acceptance of computer technology: a comparison of two theoretical models. Manag. Sci. 35 (8), 982–1003.
de Kervenoael, R., Hasan, R., Schwob, A., Goh, E., 2020. Leveraging human-robot interaction in hospitality services: incorporating the role of perceived value, empathy, and information sharing into visitors' intentions to use social robots. Tour. Manag. 78, 104042.
Dhar, R., Wertenbroch, K., 2000. Consumer choice between hedonic and utilitarian goods. J. Mark. Res. 37 (1), 60–71.
Edwards, J.K., Rosenbaum, M.S., Cheng, M., 2018. Research note: legal implications of soundalike hotels in China: haiyatt and Marvelot. Tour. Manag. 68, 162–167.
El Hedhli, K., Zourrig, H., Al Khateeb, A., Alnawas, I., 2023. Stereotyping human-like virtual influencers in retailing: does warmth prevail over competence? J. Retailing Consum. Serv. 75, 103459.
Faul, F., Erdfelder, E., Buchner, A., Lang, A.G., 2009. Statistical power analyses using G*Power 3.1: tests for correlation and regression analyses. Behav. Res. Methods 41 (4), 1149–1160.
Feine, J., Gnewuch, U., Morana, S., Maedche, A., 2019. A taxonomy of social cues for conversational agents. Int. J. Hum. Comput. Stud. 132, 138–161.
Fiske, S.T., Cuddy, A.J., Glick, P., 2007. Universal dimensions of social cognition: warmth and competence. Trends Cognit. Sci. 11 (2), 77–83.
Fiske, S.T., Cuddy, A.J., Glick, P., Xu, J., 2002. A model of (often mixed) stereotype content: competence and warmth respectively follow from perceived status and competition. J. Pers. Soc. Psychol. 82, 878–902.

Frey, C.B., Osborne, M.A., 2017. The future of employment: how susceptible are jobs to computerisation? Technol. Forecast. Soc. Change 114, 254–280.
Gao, Y.L., Mattila, A.S., 2014. Improving consumer satisfaction in green hotels: the roles of perceived warmth, perceived competence, and CSR motive. Int. J. Hospit. Manag. 42, 20–31.
Garvey, A.M., Kim, T., Duhachek, A., 2023. Bad news? Send an AI. Good news? Send a human. J. Market. 87 (1), 10–25.
Granulo, A., Fuchs, C., Puntoni, S., 2021. Preference for human (vs. robotic) labor is stronger in symbolic consumption contexts. J. Consum. Psychol. 31 (1), 72–80.
Gray, H.M., Gray, K., Wegner, D.M., 2007. Dimensions of mind perception. Science 315 (5812), 619, 619.
Gray, K., Wegner, D.M., 2012. Feeling robots and human zombies: mind perception and the uncanny valley. Cognition 125 (1), 125–130.
Gretry, A., Horváth, C., Belei, N., van Riel, A.C., 2017. 'Don't pretend to be my friend!' When an informal brand communication style backfires on social media. J. Bus. Res. 74, 77–89.
Grewal, D., Noble, S.M., Roggeveen, A.L., Nordfalt, J., 2020. The future of in-store technology. J. Acad. Market. Sci. 48, 96–113.
Güntürkün, P., Haumann, T., Mikolon, S., 2020. Disentangling the differential roles of warmth and competence judgments in customer-service provider relationships. J. Serv. Res. 23 (4), 476–503.
Gursoy, D., Chi, O.H., Lu, L., Nunkoo, R., 2019. Consumers acceptance of artificially intelligent (AI) device use in service delivery. Int. J. Inf. Manag. 49, 157–169.
Han, J.K., Sohn, Y.S., Yoo, K.W., 2015. The Korean language and the effects of its honorifics system in advertising: deferential vs. informal speech as regulatory prime on persuasive impact. Mark. Lett. 26, 321–333.
Hartman, J.B., Shim, S., Barber, B., O'Brien, M., 2006. Adolescents' utilitarian and hedonic Web consumption behavior: hierarchical influence of personal values and innovativeness. Psychol. Market. 23 (10), 813–839.
Haslam, N., 2006. Dehumanization: an integrative review. Pers. Soc. Psychol. Rev. 10 (3), 252–264.
Hayes, A., 2013. Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach. The Guilford Press, New York.
Hirschman, E.C., Holbrook, M.B., 1982. Hedonic consumption: emerging concepts, methods and propositions. J. Market. 46 (3), 92–101.
Hofmann, P., Samp, C., Urbach, N., 2020. Robotic process automation. Electron. Mark. 30 (1), 99–106.
Hou, Y., Zhang, K., Li, G., 2021. Service robots or human staff: how social crowding shapes tourist preferences. Tour. Manag. 83, 104242.
Hsieh, S.H., Lee, C.T., 2024. The AI humanness: how perceived personality builds trust and continuous usage intention. J. Prod. Brand Manag. 33 (5), 618–632.
Huang, M.H., Rust, R.T., 2018. Artificial intelligence in service. J. Serv. Res. 21 (2), 155–172.
International Federation of Robotics, 2022. World robotics 2022. https://ifr.org/news/sales-of-robots-for-the-service-sector-grew-by-37-worldwide/. (Accessed 16 January 2023).
Ivanov, S., Webster, C., 2019. Perceived appropriateness and intention to use service robots in tourism. In: Pesonen, J., Neihardt, J. (Eds.), Information and Communication Technologies in Tourism 2019: Proceedings of the International Conference in Nicosia. Springer International Publishing., Cyprus, pp. 237–248.
Jiang, Y., Wen, J., 2020. Effects of COVID-19 on hotel marketing and management: a perspective article. Int. J. Contemp. Hospit. Manag. 32 (8), 2563–2573.
Kim, Y., Kwak, S.S., Kim, M.S., 2013. Am I acceptable to you? Effect of a robot's verbal language forms on people's social distance from robots. Comput. Hum. Behav. 29 (3), 1091–1101.
Lee, K.W., Li, C.Y., 2023. It is not merely a chat: transforming chatbot affordances into dual identification and loyalty. J. Retailing Consum. Serv. 74, 103447.
Lemmink, J., Mattsson, J., 2002. Employee behavior, feelings of warmth and customer perception in service encounters. Int. J. Retail Distrib. OR Manag. 30 (1), 18–33.
Li, Z., Terfurth, L., Woller, J.P., Wiese, E., 2022. Mind the machines: applying implicit measures of mind perception to social robotics. In: 2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI), pp. 236–245.
Liu, F., Lee, Y.H., 2024. Virtually responsible? Attribution of responsibility toward human vs. virtual influencers and the mediating role of mind perception. J. Retailing Consum. Serv. 77, 103685.
Longoni, C., Bonezzi, A., Morewedge, C.K., 2019. Resistance to medical artificial intelligence. J. Consum. Res. 46 (4), 629–650.
Longoni, C., Cian, L., 2022. Artificial Intelligence in utilitarian vs. hedonic contexts: the 'Word-of-Machine' effect. J. Market. 86 (1), 91–108.
Lu, L., Cai, R., Gursoy, D., 2019. Developing and validating a service robot integration willingness scale. Int. J. Hospit. Manag. 80, 36–51.
Lv, X., Li, H., Xia, L., 2020. Effects of haptic cues on consumers' online hotel booking decisions: the mediating role of mental imagery. Tour. Manag. 77, 104025.
Lv, X., Liu, Y., Luo, J., Liu, Y., Li, C., 2021. Does a cute artificial intelligence assistant soften the blow? The impact of cuteness on customer tolerance of assistant service failure. Ann. Tourism Res. 87, 103114.
Lv, X., Shi, K., He, Y., Ji, Y., Lan, T., 2024. My colleague is not "human": will working with robots make you act more indifferently? J. Bus. Res. 176, 114585.
Marinova, D., Singh, S.K., Singh, J., 2018. Frontline problem-solving effectiveness: a dynamic analysis of verbal and nonverbal cues. J. Mark. Res. 55 (2), 178–192.
McArthur, T., 1992. The Oxford Companion to the English Language. Oxford University Press, New York.
Mende, M., Scott, M.L., Van Doorn, J., Grewal, D., Shanks, I., 2019. Service robots rising: how humanoid robots influence service experiences and elicit compensatory consumer responses. J. Mark. Res. 56 (4), 535–556.
Meuter, M.L., Ostrom, A.L., Roundtree, R.I., Bitner, M.J., 2000. Self-service technologies: understanding customer satisfaction with technology-based service encounters. J. Market. 64 (3), 50–64.
Min, H.K., Hu, Y., Ann, S., 2023. Impression management goals and job candidate's race: a test of competing models. Int. J. Hospit. Manag. 109, 103426.
Mittal, B., Lassar, W.M., 1996. The role of personalization in service encounters. J. Retailing 72 (1), 95–109.
Moore, S., Bulmer, S., Elms, J., 2022. The social significance of AI in retail on customer experience and shopping practices. J. Retailing Consum. Serv. 64, 102755.
Morera, M.D., Quiles, M.N., Correa, A.D., Delgado, N., Leyens, J.P., 2018. Perception of mind and dehumanization: human, animal, or machine? Int. J. Psychol. 53 (4), 253–260.
O'Curry, S., Strahilevitz, M., 2001. Probability and mode of acquisition effects on choices between hedonic and utilitarian options. Mark. Lett. 12 (1), 37–49.
Oikarinen, E.L., Saraniemi, S., 2016. Categorizing humorous employer brand message in a small company's online job ads. Corp. Reput. Rev. 19, 198–218.
Orth, U.R., Spielmann, N., Meyer, C., 2023. Ambient temperature in online service environments. J. Serv. Res. 26 (2), 155–172.
Patrizi, M., Šerić, M., Vernuccio, M., 2024. Hey Google, I trust you! The consequences of brand anthropomorphism in voice-based artificial intelligence contexts. J. Retailing Consum. Serv. 77, 103659.
Peng, C., van Doorn, J., Eggers, F., Wieringa, J.E., 2022. The effect of required warmth on consumer acceptance of artificial intelligence in service: the moderating role of AI-human collaboration. Int. J. Inf. Manag. 66, 102533.
Pham, H.C., Duong, C.D., Nguyen, G.K.H., 2024. What drives tourists' continuance intention to use ChatGPT for travel services? A stimulus-organism-response perspective. J. Retailing Consum. Serv. 78, 103758.
Prentice, C., Nguyen, M., 2020. Engaging and retaining customers with AI and employee service. J. Retailing Consum. Serv. 56, 102186.
Quach, S., Barari, M., Moudrý, D.V., Quach, K., 2022. Service integration in omnichannel retailing and its impact on customer experience. J. Retailing Consum. Serv. 65, 102267.
Rafiq, M., Ahmed, P.K., 1998. A customer-oriented framework for empowering service employees. J. Serv. Market. 12 (5), 379–396.
Shin, H., Perdue, R.R., Kang, J., 2019. Front desk technology innovation in hotels: a managerial perspective. Tour. Manag. 74, 310–318.
Sparks, B., Callan, V.J., 1992. Communication and the service encounter: the value of convergence. Int. J. Hospit. Manag. 11 (3), 213–224.
Tsai, W.C., Huang, Y.M., 2002. Mechanisms linking employee affective delivery and customer behavioral intentions. J. Appl. Psychol. 87 (5), 1001.
Tung, V.W.S., Law, R., 2017. The potential for tourism and hospitality experience research in human-robot interactions. Int. J. Contemp. Hospit. Manag. 29 (10), 2498–2513.
Tussyadiah, I., 2020. A review of research into automation in tourism: launching the annals of tourism research curated collection on artificial intelligence and robotics in tourism. Ann. Tourism Res. 81, 102883.
Voss, K.E., Spangenberg, E.R., Grohmann, B., 2003. Measuring the hedonic and utilitarian dimensions of consumer attitude. J. Mark. Res. 40 (3), 310–320.
Wang, Z., Mao, H., Li, Y.J., Liu, F., 2017. Smile big or not? Effects of smile intensity on perceptions of warmth and competence. J. Consum. Res. 43 (5), 787–805.
Wirtz, J., Patterson, P.G., Kunz, W.H., Gruber, T., Lu, V.N., Paluch, S., Martins, A., 2018. Brave new world: service robots in the frontline. J. Serv. Manag. 29 (5), 907–931.
Wong, I.A., Yang, F.X., 2020. A quarantined lodging stay: the buffering effect of service quality. Int. J. Hospit. Manag. 91, 102655.
Wu, L., Shen, H., Fan, A., Mattila, A.S., 2017. The impact of language style on consumers' reactions to online reviews. Tour. Manag. 59, 590–596.
Yam, K.C., Tang, P.M., Jackson, J.C., Su, R., Gray, K., 2022. The rise of robots increases job insecurity and maladaptive workplace behaviors: multimethod evidence. J. Appl. Psychol. 108 (5), 850–870.
Yim, Y.-C., Yoo, S.-C., Sauer, P.L., Seo, J.H., 2014. Hedonic shopping motivation and co-shopper influence on utilitarian grocery shopping in superstores. J. Acad. Market. Sci. 42 (5), 528–544.
Yu, C.E., Ngan, H.F.B., 2019. The power of head tilts: gender and cultural differences of perceived human vs human-like robot smile in service. Tour. Rev. 74 (3), 428–442.
Zhao, L., Zhang, M., Ming, Y., Niu, T., Wang, Y., 2023. The effect of image richness on customer engagement: evidence from Sina Weibo. J. Bus. Res. 154, 113307.
Zhou, P., Xie, Y., Liang, C., 2023. How to increase consumers' continued use intention of artificial intelligence voice assistants? The role of anthropomorphic features. Electron. Mark. 33, 60.